

EDUCATION

Lovely Professional University
Bachelor of Technology (Computer Science & Engineering)

Jalandhar, India
July 2022 - May 2026

SKILLS SUMMARY

- **Programming:** Python, SQL, NumPy, Pandas
- **AI/ML:** Machine Learning, Deep Learning, PyTorch, NLP, ML Pipelines, Model Evaluation
- **Generative AI:** LLMs, Generative AI, RAG, Prompt Engineering, Embeddings, Vector Search, Context Engineering
- **Agentic AI:** AI Agents, LangGraph, Multi-Agent Workflows, Tool Use, Workflow Orchestration
- **AI Systems:** Semantic Search, Model Serving, AI Evaluation, Monitoring, **Observability**
- **Frameworks:** LangChain, FastAPI
- **Engineering:** Performance Optimization, Debugging, REST APIs, Docker, Git, GitHub

WORK EXPERIENCE

- AI Intern | Know AI | [LINK](#)

June 2026 - August 2026

 - Built **Generative AI applications** using LLM inference, NLP, prompt engineering, and model evaluation.
 - Developed **Python AI services and REST APIs** integrating LLMs, search, and automation workflows.
 - Designed **RAG pipelines** using embeddings, vector databases, and semantic search for knowledge-grounded retrieval.
 - Developed AI/ML **evaluation pipelines** for model testing, error analysis, output validation, and performance analysis.
 - Optimized AI workflows through **debugging, monitoring, observability, evaluation, and performance analysis**.
- Data Engineer Intern | flickDONE | [LINK](#)

June 2025 - August 2025

 - Built **Python-based ML data pipelines** processing 50,000+ records for preprocessing, training, and validation.
 - Engineered automated **data preprocessing and validation workflows**, reducing manual verification by 70%.
 - Optimized AWS ML pipelines, improving **execution speed by 25%** through bottleneck analysis and workflow optimization.
 - Developed reusable AI workflows for **document processing, conversational AI, and business automation**.
 - Collaborated with engineering teams to improve **scalability, reliability, and maintainability** of ML platforms.

PROJECTS

- LangGraph Workflow Orchestrator | [LIVE](#)

August 2025

 - Built a stateful **Agentic AI platform** using Python, LangGraph, LangChain, FastAPI, and Gemini.
 - Designed multi-step **LLM workflows** with state management, conditional routing, and error recovery.
 - Developed modular components for **task decomposition, tool use, and agent coordination**.
 - Exposed AI workflows through **FastAPI REST APIs**, enabling scalable AI task automation.
 - Improved workflow reliability through **checkpointing, testing, failure handling, and monitoring**.
- Production-Grade RAG Engine | [LIVE](#)

June 2024 - August 2024

 - Built an end-to-end **RAG application** using Python, FastAPI, LangChain, Qdrant, and AWS.
 - Developed **document ingestion, chunking, embedding, retrieval, and context generation** pipelines.
 - Implemented **embedding-based semantic search** with Qdrant for efficient document context retrieval.
 - Added **reranking and retrieval evaluation** to improve context relevance and response quality.
 - Improved AI service **reliability and maintainability** through testing, monitoring, fallback execution, and error handling.
- Regulatory Compliance RAG Assistant | [LIVE](#)

April 2026 - June 2026

 - Built a Generative AI application using Python, Streamlit, LangChain, embeddings, vector search, and LLMs.
 - Designed document ingestion and **semantic retrieval pipelines** for regulatory compliance analysis.
 - Implemented **knowledge-grounded generation** to deliver relevant and reliable compliance responses.
 - Developed interactive document upload and querying workflows using **Python and Streamlit**.
 - Improved **GenAI reliability** through retrieval testing, evaluation, debugging, monitoring, and quality checks.

CERTIFICATES

- Python, Data Science & ML (Cipherschools) | [CERTIFICATE](#)

August 2024

 - Built machine learning models using PyTorch and applied ML techniques for model development and evaluation.
 - Developed **Dockerized environments** supporting reproducible experimentation and deployment
- Generative AI with Large Language Models (Coursera) | [CERTIFICATE](#)

March 2024

 - Applied prompt engineering techniques for production LLM applications.
 - Built **retrieval workflows, AI output evaluation, and RAG pipeline** performance validation.